Comparative Analysis of Machine Learning Models for Weather Forecasting in Porto Alegre, RS

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Introduction

Machine Learning (ML) has emerged as a robust framework for predicting and characterizing complex physical phenomena. By leveraging high-dimensional datasets, these models can capture non-linear dynamics and identify anomalous behaviors, facilitating the development of predictive tools for environmental risk assessment[1]. This is particularly critical for forecasting extreme climate events, where accurate early warnings are essential for mitigating the socio-economic impacts of natural disasters.

This study evaluates the performance of ML architectures in forecasting daily maximum and mean temperatures for the Porto Alegre metropolitan area—a region increasingly vulnerable to climate extremes. Utilizing spatially aggregated data from the ERA5 reanalysis, we implemented time-series forecasting across varying lags and lead times. Our results indicate that different model architectures yield comparable performance, with no statistically significant differences relative to the expected error. Furthermore, we investigate the emergence of "error plateaus," offering insights into the inherent predictability limits of the local atmospheric system.

Research Objectives

The primary goal of this research is to evaluate the performance of diverse Machine Learning regression architectures as predictive committor functions for localized weather forecasting, specifically targeting daily temperature extremes (max and mean).

Specific Objectives

- Develop and optimize multivariate forecasting models to predict maximum and mean temperatures for a 10-day lead horizon.
- Quantify model sensitivity and predictive performance across varying temporal lags to identify optimal memory windows for regional atmospheric dynamics.
- Comparative analysis of gradient-boosted trees (XGBoost) and recurrent architectures (LSTM) in capturing localized temperature variability.

Methodology

The forecasting framework was designed to compare the predictive efficiency of tree-based ensemble methods against recurrent neural architectures. We implemented a multi-step forecasting approach to predict maximum (T_{max}) and mean (T_{mean}) temperatures for a 10-day lead horizon.

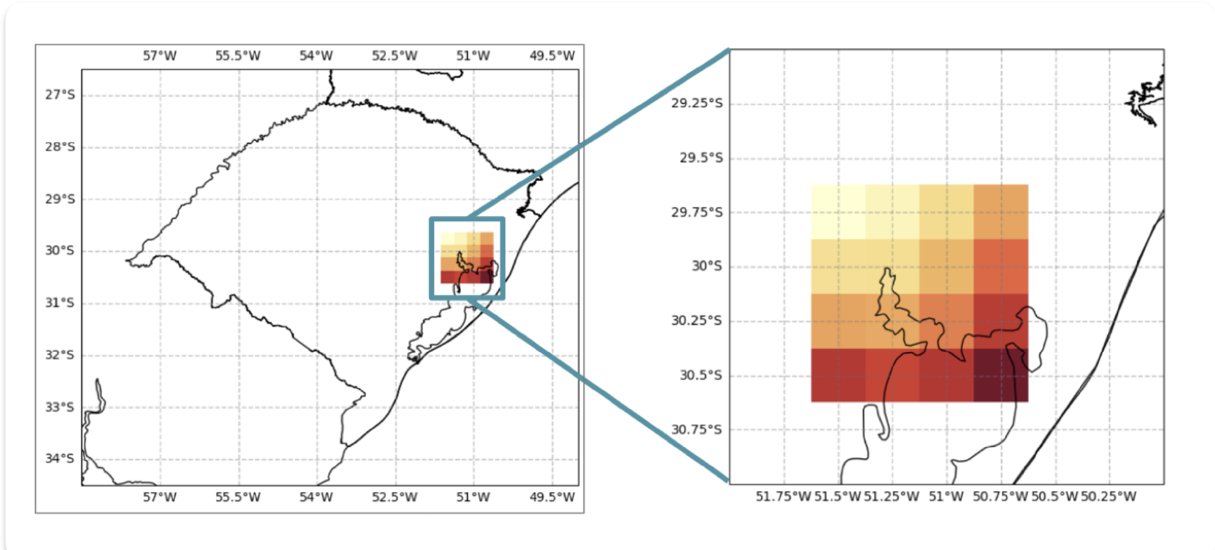
Data Acquisition and Preprocessing

- Dataset: Daily atmospheric data from the ERA5 reanalysis (1990–1999).
- Spatial Aggregation: Features were extracted from a 4x4 grid centered over the Porto Alegre metropolitan region. Statistical features (mean, max, min, standard deviation and variance) were computed to capture sub-grid variability.
- Feature Selection: Univariate: Historical T_{max} and T_{mean} time series. Multivariate: Inclusion of geopotential, sea level pressure, and soil moisture.
- Data Partitioning: The dataset was split chronologically: 1990–1997 for training/validation and 1998–1999 for out-of-sample testing.

Experimental Setup

We evaluated model sensitivity to historical "memory" by varying the temporal lags: $\tau \in \{5, 15, 30, 60, 90\}$ days.

- XGBoost (eXtreme Gradient Boosting): Utilized for its efficiency in handling tabular data and capturing non-linear feature interactions through gradient-boosted decision trees.
- LSTM (Long Short-Term Memory): A recurrent architecture employed to capture long-range temporal dependencies and sequential patterns inherent in atmospheric dynamics.



Evaluation Metrics

The predictive performance was quantified using the Mean Absolute Error (MAE) and the Coefficient of Determination (R^2):

$$MAE = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|^2$$

$$R2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)}{\sum_{i=1}^{N}(y_i - \bar{y})}$$

where y_i represents the observed values, $\hat{y}_i$ the predicted values, and $\bar{y}$ the mean of the observations.

Results

The models exhibit a characteristic decay in predictive accuracy as the forecast horizon (lead time) increases. Despite this, the results for a 24-hour lead time (1 day) align closely with established "Good Error Range" benchmarks found in the literature: from 0.8°C to 1.2°C for T_{mean} and from 1.5°C to 2.5°C for T_{max} [2].

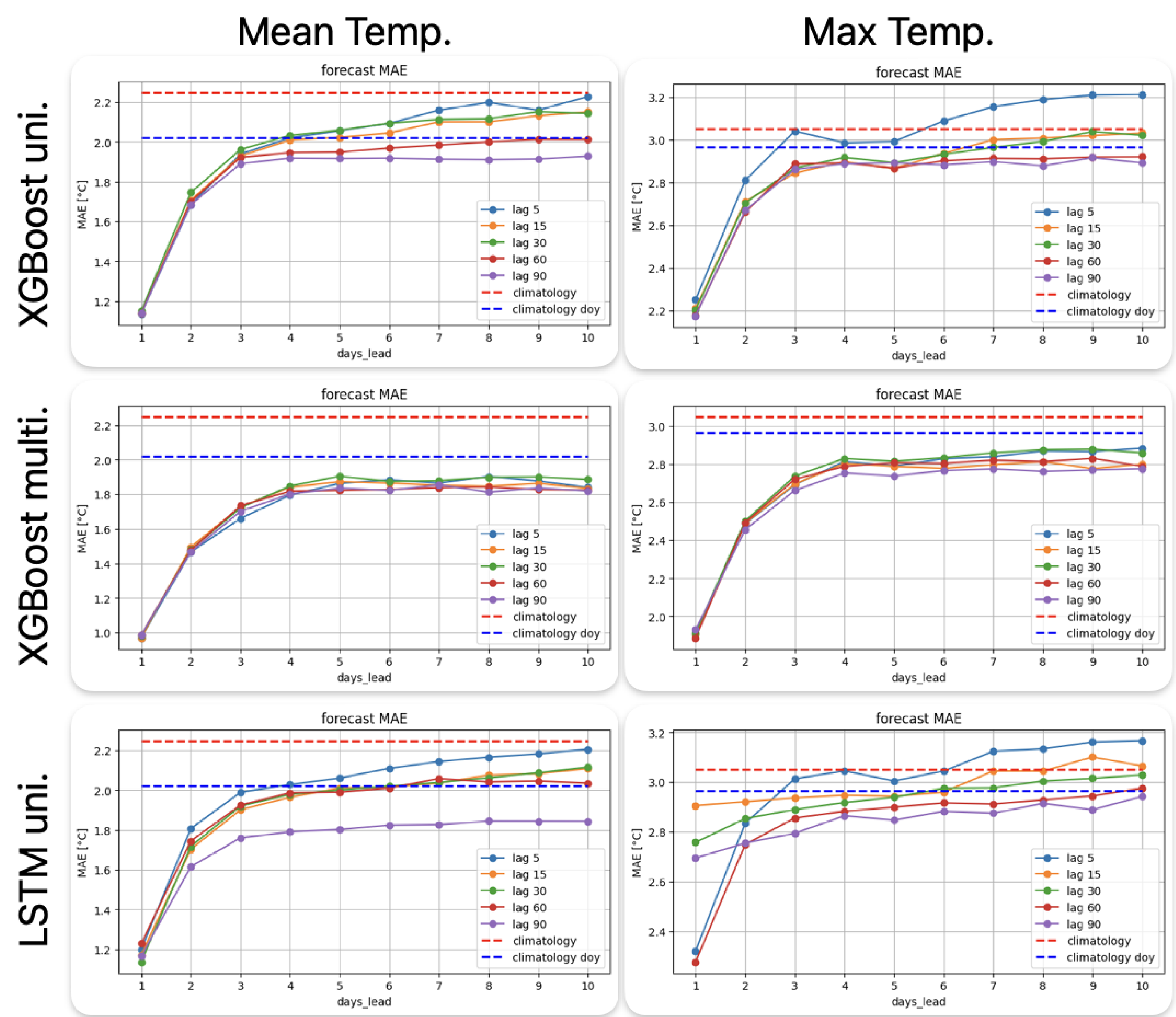


Figure 1: MAE as a function of lead days across diverse lag configurations and architectures.

A visual analysis indicates that the choice of temporal lag ($\tau \in \{5, \ldots, 90\}$ days) has a marginal impact on performance. This suggests that the predictive skill is primarily driven by immediate local atmospheric states rather than long-term historical memory.

Error Dynamics and the "MAE Plateau"

As illustrated in Figure1, the Mean Absolute Error exhibits a rapid increase during the first three days but stabilizes into a plateau after lead day 4. This behavior is further elucidated in Figure2, which compares the observed vs. predicted time series for lead days of 1 and 10.

The emergence of this plateau suggests a fundamental transition in model behavior:

1. Short-term Dynamics: For lead days less than 4, the architectures effectively capture stochastic local fluctuations and transient weather patterns.
2. Long-term Convergence: As the forecast horizon extends, the models converge toward a climatological mean behavior for month and day-of-year aggregation. The "plateau" reflects the model's transition from specific event prediction to a regression-to-the-mean strategy, indicating an inherent predictability limit for the selected spatial resolution and feature set.

A notable observation regarding to multivariate models is that the forecast plateau remains below the expected climatological baseline. This suggests that the models are capturing complex seasonal aggregations within the atmospheric variables, which enhances predictive performance beyond simple historical averages.

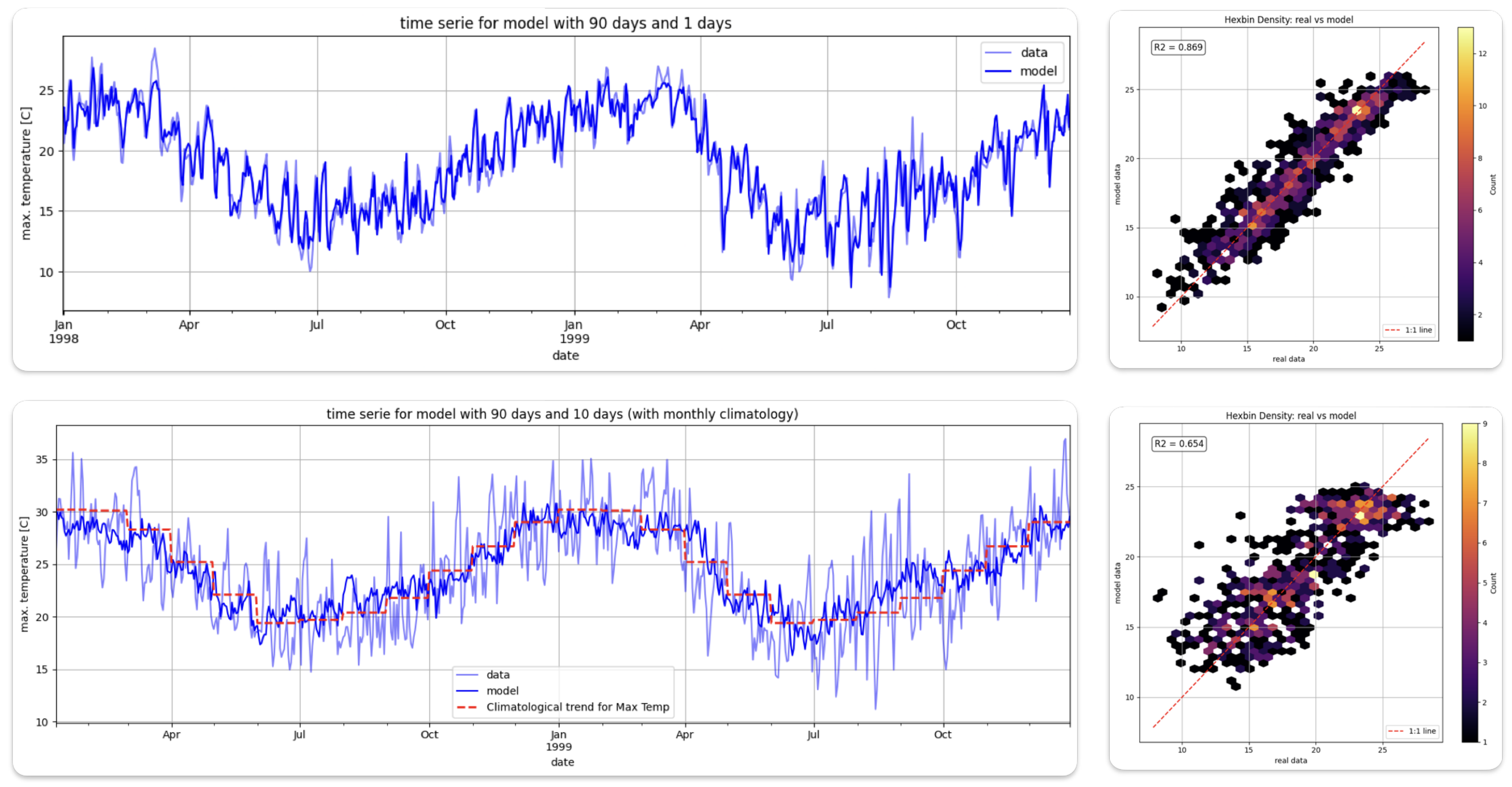


Figure 2: Observed vs. Predicted T_{max} (XGBoost, 90-day lag) for Lead 1 (top) and Lead 10 (bottom).

Conclusion

This research evaluates the efficacy of Machine Learning (ML) architectures (XGBoost and LSTM) as predictive committor functions for forecasting temperature extremes in Porto Alegre, RS, utilizing statistically aggregated features from ERA5 reanalysis maps. Our results demonstrate that while 24-hour lead-time errors (one day) align closely with established literature benchmarks, predictive performance exhibits a characteristic error plateau after lead day 4, signaling a transition from resolved stochastic dynamics to a regression toward climatological mean behavior. The study concludes that the integration of multivariate spatial data is the primary driver of forecast accuracy, outperforming specific architecture selection and providing a robust framework for identifying the inherent predictability limits of localized atmospheric systems.

References

[1] Francisco A. Rodrigues (2023) *EPL* 144 22001 doi: 10.1209/0295-5075/ad0575.

[2] WWRP/WGNE Joint Working Group on Forecast Verification Research https://wmo.int/wwrpwgne-joint-working-group-forecast-verification-research